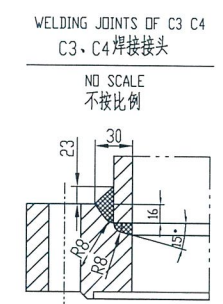
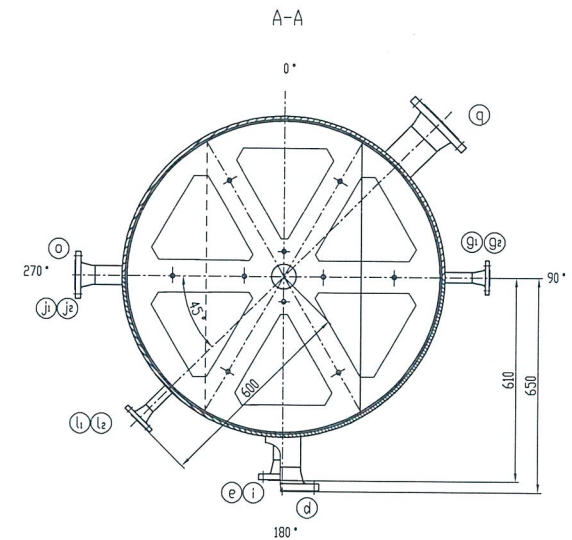
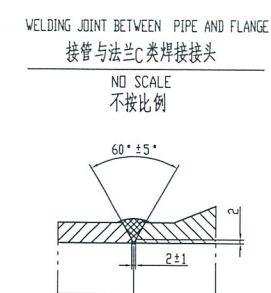
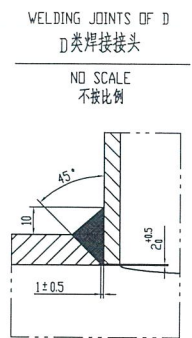
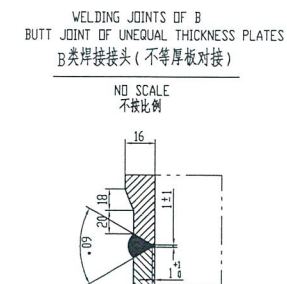
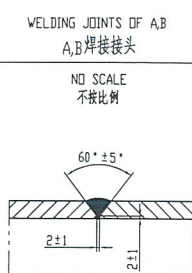
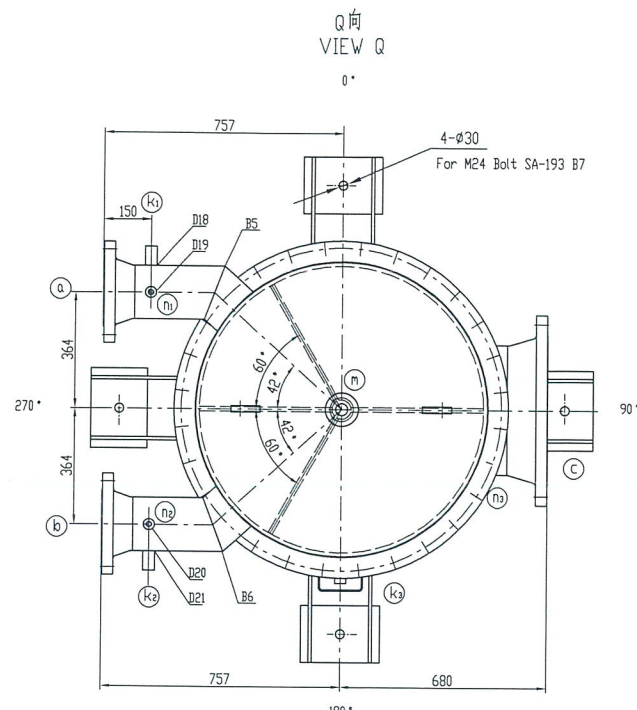


NO. VERIFIKASI: JKKP/2025/01-05/T-01/00914
VERIFICATION NO.

TARIKH: 04/08/2025
DATE



DESIGN SPECIFICATION 设计数据表					
DESIGN AND MANUFACTURE CODE 设计和制造规范		ASME CODE SECTION VIII, DIV.1, 2023ED factories and machinery (steam boiler and unfired pressure vessel) regulations, 1970			
SHELL SIDE 壳程	TUBE SIDE 管程	SEISMIC LOAD 抗震载荷	N/A		
OPERATING PRESSURE 工作压力 MPa.G	0.0755(a)	WIND LOAD 风荷载	N/A		
OPERATING PRESSURE 设计压力 MPa.G	0.1	EXPECTED DESIGN SERVICE TIME (YEAR) 预期设计使用寿命 (年)	20		
M.A.W.P 最大允许工作压力 MPa.G	0.1099 *	CERTIFICATION MARK 认证标志	DOSH		
OPERATING TEMPERATURE 工作温度 °C	90.24	MS REGISTRATION 制造注册			
DESIGN TEMPERATURE 设计温度 °C	99	WELDING CODE 焊接规范	ACC. TO WPS AND PQR 依据 WPS 和 PQR		
最低设计金属温度 °C		WELDING STRUCTURE 焊接结构	FULL PENETRATION WELD EXCEPT NOTED 除注明外均为全透焊接		
HEAT MEDIUM (IN/OUT) 介质 (进出口)	0 °C AT 0.1 MPa 蒸汽/水	WELDING TYPE 确定焊接类型	ACC. TO THINNESS PLATE EXCEPT NOTED 按板厚规范除注明外		
MEDIUM CHARACTERISTIC 介质特征	NON-LETHAL 非致命	WELDING RDT TYPE 焊接 RDT 类型	ACC. TO WPS AND PQR 依据 WPS 和 PQR		
CORROSION ALLOWANCE 腐蚀裕量	1.0	WELDER JENET CATEGORY 焊接接头类别	EX. METHOD AND AS PER 方法与数据		
JOINT COEFFICIENT 焊接接头系数	0.85				
TOTAL VOLUME 总容积 m³	2.42	N.D.E 无损检测	SPOT RT, AS PER UW-52		
HEAT TRANSFER AREA 换热面积 m²	150				
TUBE DIMENSION 管径规格 mm	Ø16X0.9/1.2				
NO. OF TUBE PASSES 管程数	1				
IMPACT TEST 冲击试验	NOT REQUIRED AS PER UCS-66, UG68/FALHAST 或 UCS-66 要求, UG68/FALHAST, 不需要	P.W.H.T 焊后热处理			
SAFETY VALVE ARRANGEMENT 安全阀设置	-	TUBE SIDE 壳程	NOT REQUIRED AS PER UCS-56 EXCEPT WELDER CHANNEL 或 UCS-56 要求, 不要求 (焊后热处理)		
SET PRESSURE OF SAFETY VALVE 安全阀设定压力 MPa.G	-	SHELL SIDE 管程	NOT REQUIRED AS PER UCS-56 或 UCS-56 要求, 不要求		
HYDRAULIC TEST PRESSURE 水压试验压力 MPa.G	0.2	PAINTING, PACKING AND TRANSPORT 涂装、包装、运输	AS PER THE TECHNICAL REQUIREMENTS 按规范技术要求		
HYDRAULIC TEST TEMPERATURE 水压试验温度 °C	10	NOZZLES AND SUPPORTS ORIENTATION 开口及支撑方位	AS PER THIS DWG. 按此图		
GAS LEAKAGE TEST PRESSURE 气密性试验压力 bar.G	-	MANUAL POSITION 操作位置	AS PER THIS DWG. 按此图		
		NET MASS 设备重量 (kg)	3955		

NOZZLE TABLE 管口表

NOZZLE 管口	CLASS 压力等级	NPS 公称管径	FLANGE CODE 连接法兰标准	CONT. TYPE 连接形式	USE 用途	REMARKS 备注
a	300	DN150 6"	ASME B16.5-2020	WN RF	FEEDWATER INLET 给水进口	
b	300	DN150 6"	ASME B16.5-2020	WN RF	FEEDWATER OUTLET 给水出口	
c	150	DN63.5 1 1/2"	ASME B16.5-2020	WN RF	STEAM INLET 蒸汽进口	
d	150	DN80 3"	ASME B16.5-2020	WN RF	DRAIN TAP NOZZLE 旁路排水入口	
e	150	DN50 2"	ASME B16.5-2020	WN RF	VENT NOZZLE AT STEEL SIDE 旁侧排气口	
G1-2	150	DN25 1"	ASME B16.5-2020	WN RF	LEVEL GAUGE NOZZLE 液位计接口	
h	150	DN80 3"	ASME B16.5-2020	WN RF	DRAIN NOZZLE 疏水口	
i		DN50 2"		WELDING 焊接连接	SIGMAZ NOZZLE 两相流信号口	
J1-2	150	DN25 1"	ASME B16.5-2020	WN RF	MAGNETIC LEVEL GAUGE NOZZLE 磁翻板液位计接口	
K1-2		M20×1.5		INNER SCREW THREAD 内螺纹	PRESSURE GAUGE NOZZLE 压力表接口	
L1-3	150	DN25 1"	ASME B16.5-2020	WN RF	REMOTE THROTTLED MAGNETIC FLAP LEVEL GAUGE 远传带磁翻板液位计接口	
M		M24×1.5		INNER SCREW THREAD 内螺纹	VENT NOZZLE OF WATER CHAMBER 水室排气口	
N1-3		M27×2		INNER SCREW THREAD 内螺纹	THERMOMETER NOZZLE 温度计接口	
O	150	DN50 2"	ASME B16.5-2020	WN RF	W NOZZLE 汽水保护接口	
Q	150	DN80 3"	ASME B16.5-2020	WN RF	EMERGENCY DRAIN NOZZLE 紧急疏水口	

THE MATERIAL SPECIFICATION INCLUDING THE GRADE, TYPE, AND CLASS IF APPLICABLE OF MAIN PRESSURE PARTS

主要受压元件材料标准, 包括等级、类型、级别 (如果有的话)

THE ELEMENT NAME 元件名称	THE SUBASSEMBLY WHICH IT BELONGS TO 所属部件	THE MATERIAL SPECIFICATION 材料标准	THE ELEMENT NAME 元件名称	THE SUBASSEMBLY WHICH IT BELONGS TO 所属部件	THE MATERIAL SPECIFICATION 材料标准
ELLIPTICAL HEAD 椭圆形头	WATER DAWM 水室	SA-516M Gr.485	ELLIPTICAL HEAD 椭圆形头	WATER DAWM 水室	SA-516M Gr.485
SHELL OF WATER CHAMBER 水室筒身	WATER DAWM 水室	SA-516M Gr.485	ELLIPTICAL HEAD 椭圆形头	WATER DAWM 水室	SA-516M Gr.485
NEZZLE 喷嘴	WATER DAWM 水室	SA-106M Gr.B	NEZZLE 喷嘴	WATER DAWM 水室	SA-106M Gr.B
TURNKEY 管架	TURBINE HANDLE 涡轮机	SA-266M Gr.2	TURNKEY 接管法兰	TURBINE HANDLE 涡轮机	SA-105M
U-TUBE U-型管	TURBINE HANDLE 管架	SA-688M TP304			

TECHNICAL REQUIREMENTS

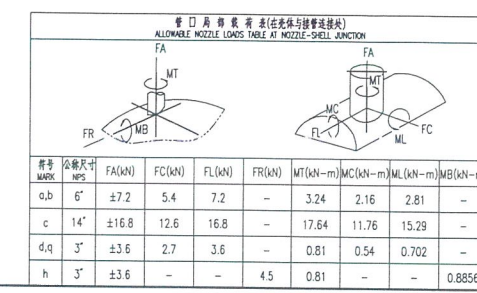
- 1 THE PERPENDICULARITY TOLERANCE OF TUBESHEET SEAL SURFACE AND SHELL AXIS IS 2mm.
- 2 TUBE-TO-TUBESHEET JOINT IS STRENGTH WELDING .
- 3 LIFTING LUG IN WATER CHAMBER HEAD IS ONLY USED TO LIFT THE WATER CHAMBER, NOT THE WHOLE EQUIPMENT.
- 4 AFTER ASSEMBLY OF SHELL AND TUBE BUNDLES, PERFORM HYDROTEST AT SHELL SIDE. THE TUBE END SHOULD BE FREE OF LEAKAGE. AFTER THE TEST IS ACCEPTED, PERFORM HYDROTEST AT TUBE SIDE.
- 5 PERFORM MT TO WELDING OF THE BACKING PLATE OF SADDLE, IN ACCORDANCE WITH ASME VIII DIV.I APPENDIX 6.
- 6 BEFORE DELIVERY, CAP EACH NOZZLE. FILL IN NITROGEN OF 0.15MPa. THE NITROGEN FILLING PRESSURE SHALL NOT BE MORE THAN 0.1MPa BEFORE DELIVERY.

技术要求

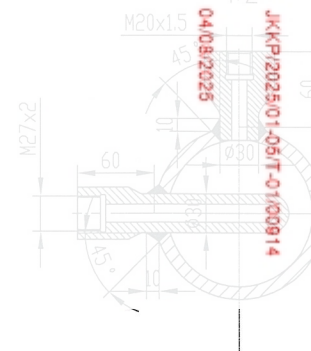
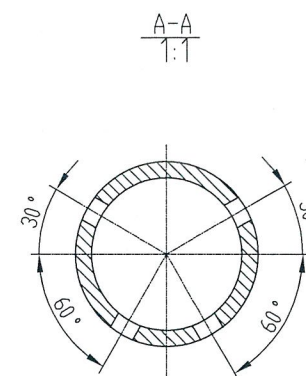
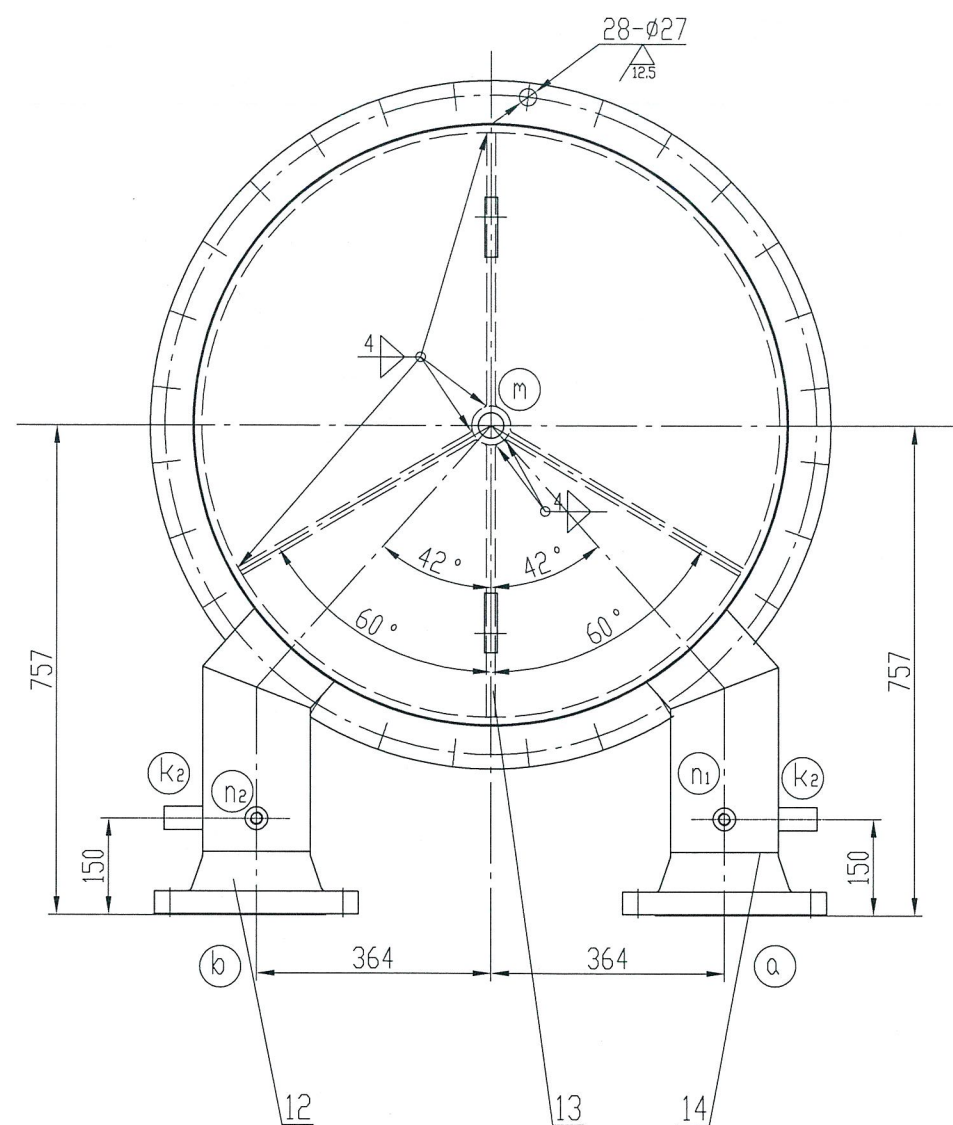
1. 管板密封面与壳体轴线垂直度公差为 2mm 。
2. 管子与管板的连接采用强度焊。
3. 水箱封头上的吊耳只用于起吊水箱，不能用于起吊整个设备。
4. 管系与壳体组装完成后进行水压试验，并检查管头不得渗漏，最后进行管程水压试验。
5. 耳式连接板与筒体的焊缝进行100%磁粉检测，符合ASME第Ⅲ卷第11附录表A要求。
6. 设备出厂前各管口须封装，并充氮保护，充氮压力为 15MPa ，出厂前不得低于 0.1MPa 。
7. 安全附件设置在管路上。

☐ inspected ☒ reviewed ☐ witnessed

June. 18, 2024



7	TXNX23-023.02	SHELL 外壳		1	COMPONENT PARTS 组合件	1596			
6	TXNX23-020.04	GASKET OF SHELL SIDE 壳侧垫片		1	PTFE	0.1		REFER TO 图月	
5	TXNX23-023.01	TUBE BUNDLE 管束		1	COMPONENT PARTS 组合件	1765			
4	TXNX23-020.02	GASKET OF WATER SIDE 水侧垫片		1	PTFE	0.2		REFER TO 图月	
3	ASME B18.2.2	NUTS 螺母 M24		56	SA-194M 2H	0.05	2.4		
2	ASME B18.2.1	FULL SCREW BOLTS 全螺纹螺栓 M24x270		28	SA-193M B7	0.73	17.52		
1	TXNX23-020.01	WATER CHAMBER 水室		1	COMPONENT PARTS 组合件	573		REFER TO 图月	
件号 ITEM	DRAW NO. 图号或标准	DESCRIPTION 名称	QTY. 数量	MATERIAL 材料	UNIT 单位	TOTAL 总量	NOTE 备注		
						WEIGHT (kg) 重量(kg)			
			LOW PRESSURE HEATER			TXNX23-023.00			
			低压加热器			A23-D023			
			JD-150			Z24101.82.62-1			
LABEL 标记	PLACE 位置	SUBAREA 分区	NO/OF DOCUMENT 页码/文件号	TECHNICAL 技术	DATE 年月日	RENEWABLE ENERGY POWER PLANT PHASE I PROJECT IN SELANGOR STATE, MALAYSIA 马来西亚雪兰莪再生能源电站二期项目			
DESIGN 设计									
COLLATION 校对									
CHECK 审核									
APPROVE 批准									
			REVISION NO. 版本	QTY. 数量	WEIGHT 重量	PROPORTION 比例	 南京汽轮电机集团泰兴宁兴机械有限公司 NINGJI TURBINE ELECTRIC MACHINERY GROUP TAI XING NINGXING MACHINERY CO., LTD.		
			0	1	3955	1:12			
			PAGE OF 共第 张						

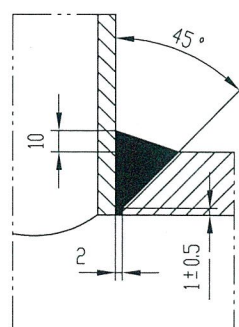


2020.6.18

				WATER CHANNEL				TXNX23-022.01			
				水室							
MARK 标记	QTY 外数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日						
DESIGNED BY 设计	[Signature]		2014.6.14				MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	
CHECKED BY 校对	[Signature]		2014.6.14				ASSEMBLY PARTS 组合件	1	573	1:8	
REVIEWED BY 审核							 南京汽轮机集团 泰兴宁兴机械有限公司 NANJING TURBINE & ELECTRIC MACHINERY GROUP TAIXING NINGXING MACHINERY CO., LTD.				
APPROVED BY 批准											
						PAGE OF					
						共 张 第 张					

Technical drawing of a mechanical part showing a cross-section. The drawing includes the following dimensions and features:

- Overall width: 23
- Overall height: 30
- Radius of the top-left corner: $R8$
- Radius of the bottom-left corner: $R8$
- Radius of the bottom-right corner: $R8$
- Angle of the bottom-right corner: 15°
- Dimension 16 indicates the distance from the bottom edge to the center of the bottom-right corner.



Technical drawing of a mechanical part with dimensions and tolerances. The drawing shows a cross-section of a part with a central hole. The dimensions and tolerances are as follows:

- Top horizontal dimension: 16
- Left vertical dimension: 18
- Left vertical dimension: 20
- Left circular dimension: $\phi 09$
- Right vertical dimension: 1 ± 1
- Right vertical dimension: $1 \begin{smallmatrix} +1 \\ 0 \end{smallmatrix}$
- Bottom horizontal dimension: 10

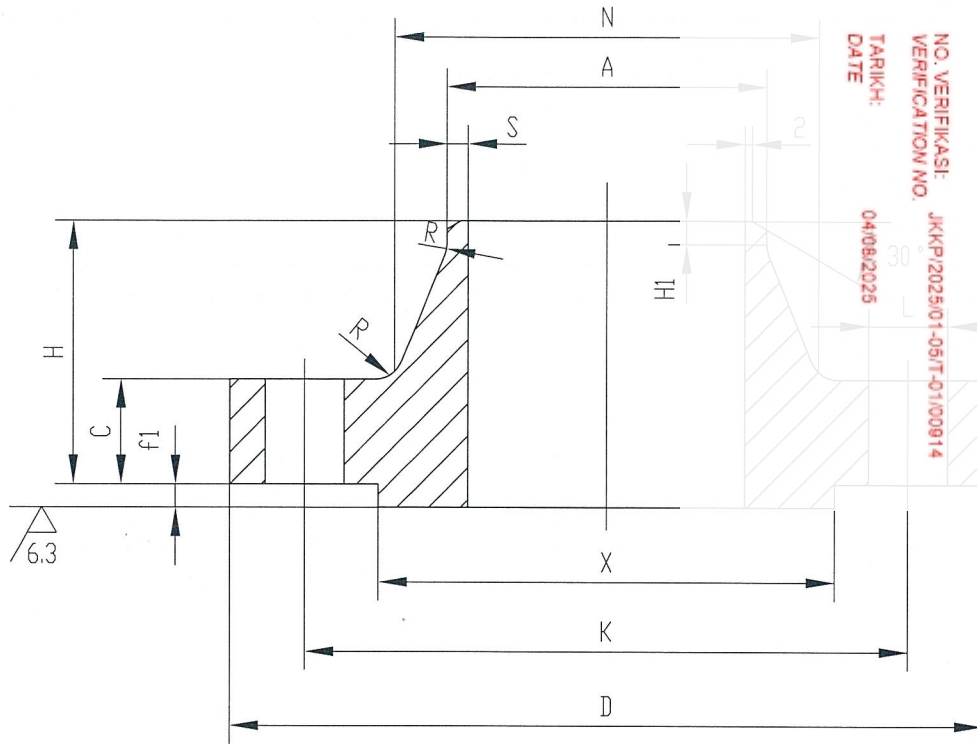


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DATE



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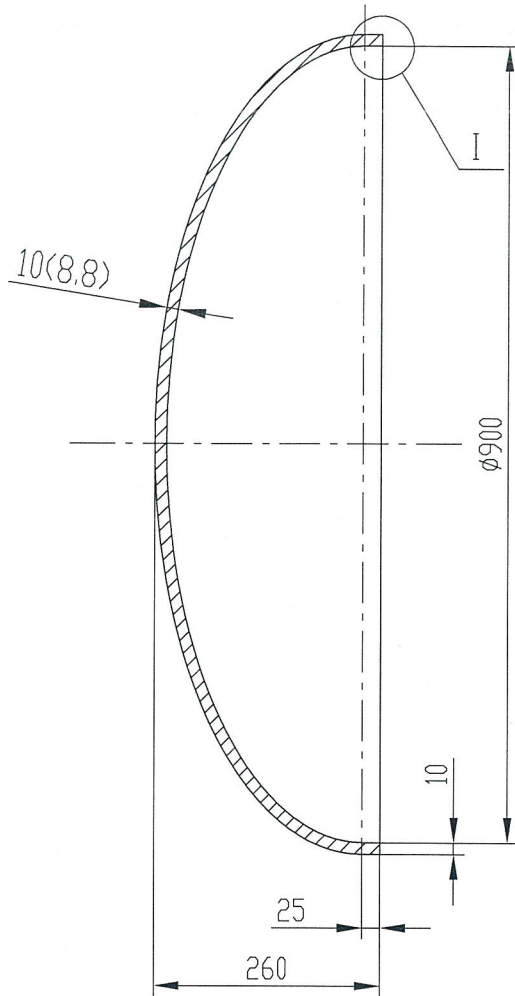
☐ Inspected ☒ Reviewed ☐ Witnessed

2024.6.18

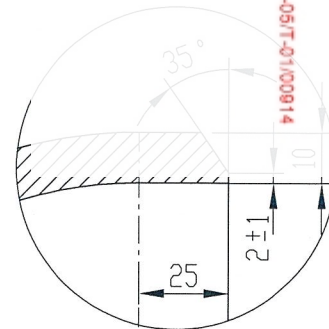
ASME B16.5 WN RF

MODEL 型号	A	D	K	L	NL OF SCREW HOLES 螺孔数	X	f1	C	H	S	N	R	NL OF FLANGE 法兰数	H1	UNIT WEIGHT 单件重	MATERIAL 材料
class300 6"	168.3	318	269.7	7/8"	12	215.9	2	35	97	10.97	206	13	2	16	12	SA-105M

						FLANGE 法兰				TXNX23-020.01.01	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	
DESIGNED BY 设计	2020.6.14										
CHECKED BY 校对	2024.6.4										
REVIEWED BY 审核											
APPROVED BY 批准						PAGE OF 共 张 第 张					



DETAIL DRAWING OF SECTION
I 部详图



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DATE

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NOTES

技术要求

- 1.THE PLATE USED FOR HEAD SHALL BE SUPPLIED IN A ROLLED STATE AND SHALL COMPLY WITH SA-516 SPECIFICATION IN PART A OF ASME SECTION II; 封头用板材以轧制状态供货, 且应符合ASME第II卷A篇SA-516规范的所有相关要求;
- 2.THIS HEAD IS A 2:1 STANDARD ELLIPTICAL HEAD AND SHALL BE MANUFACTURED IN ACCORDANCE WITH ASME SECTION VIII; 此封头为2:1的标准椭圆封头, 应按照ASME第VIII卷有关各篇制造方面的要求制造;
- 3.THE MINIMUM THICKNESS AFTER HEAD FORMING SHALL NOT BE LESS THAN 8.8MM. 封头成形后最小厚度不得小于8.8MM.

TÜVRheinland
☐ inspected ☒ reviewed ☐ witnessed
2024.6.14

						HEAD 封头 900x10(8.8)				TXNX23-020.01.05					
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日										
DESIGNED BY 设计	陈伟		2024.6.14		MATERIAL 材料		QTY 数量	WEIGHT 重量	PROPORTION 比例						
CHECKED BY 校对	徐子		2024.6.14												
REVIEWED BY 审核					SA-516M.Gr.485		1	74.1		TAIXING NINGXING MACHINERY CO., LTD.					
APPROVED BY 批准							PAGE OF 共 张 第 张			泰兴宁兴机械有限公司					

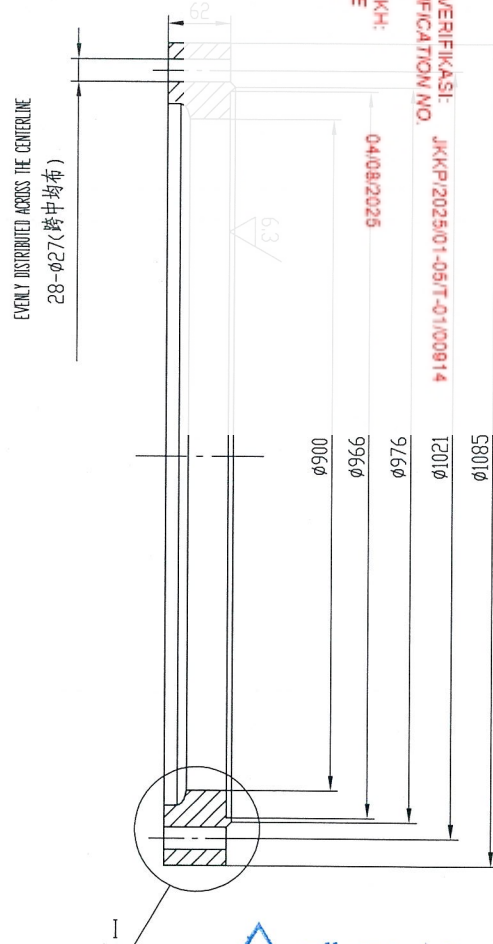


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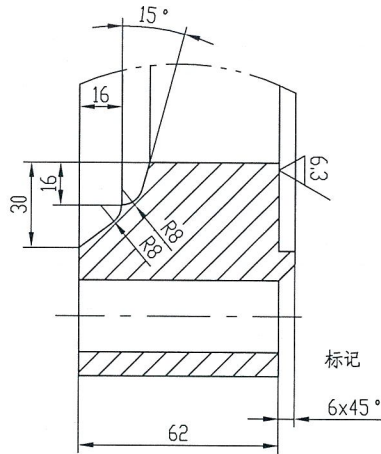


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Document Verification

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DETAIL DRAWING OF SECTION I
I 部详图
NO SCALE
不按比例

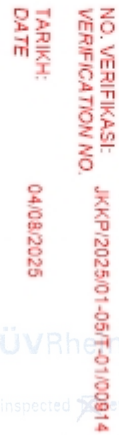


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☐ inspected ☒ reviewed ☐ witnessed
2024.6.18

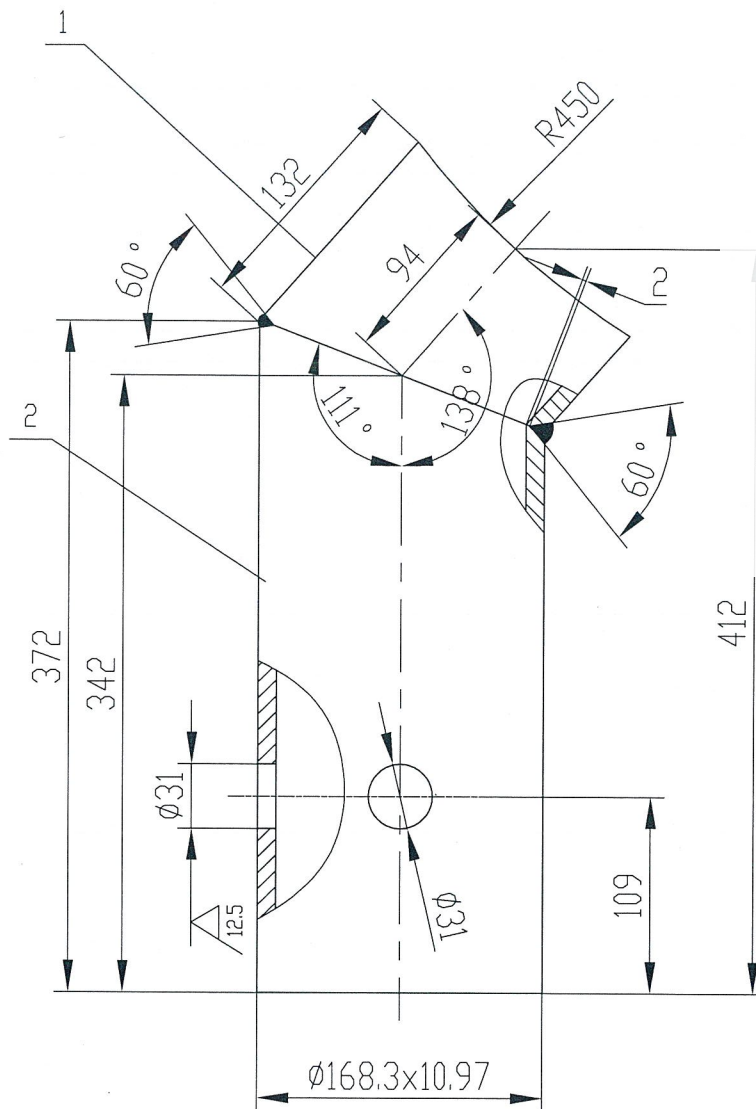
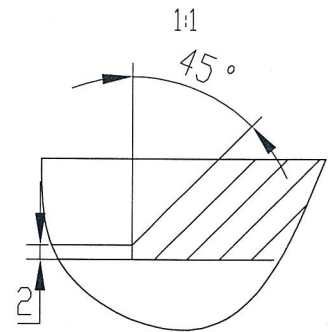
NOTES
技术要求

1.THE FLANGE MATERIAL SHALL BE SUPPLIED IN A NORMALIZED STATE
AND SHALL COMPLY WITH ASME SECTION II, PART A.
法兰材料以正火状态供货，且应符合ASME第II卷A篇SA-105M规范的所有相关要求。

						GIRTH FLANGE 设备法兰				TXNX23-020.01.07					
														TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例						
DESIGNED BY 设计	陈伟		2024.6.14			SA-105M	1	110	1:2						
CHECKED BY 校对	徐灯		2024.6.14			PAGE OF 共 张 第 张									
REVIEWED BY 审核															
APPROVED BY 批准															



DRAWING OF OPENING GROOVE
开孔坡口图



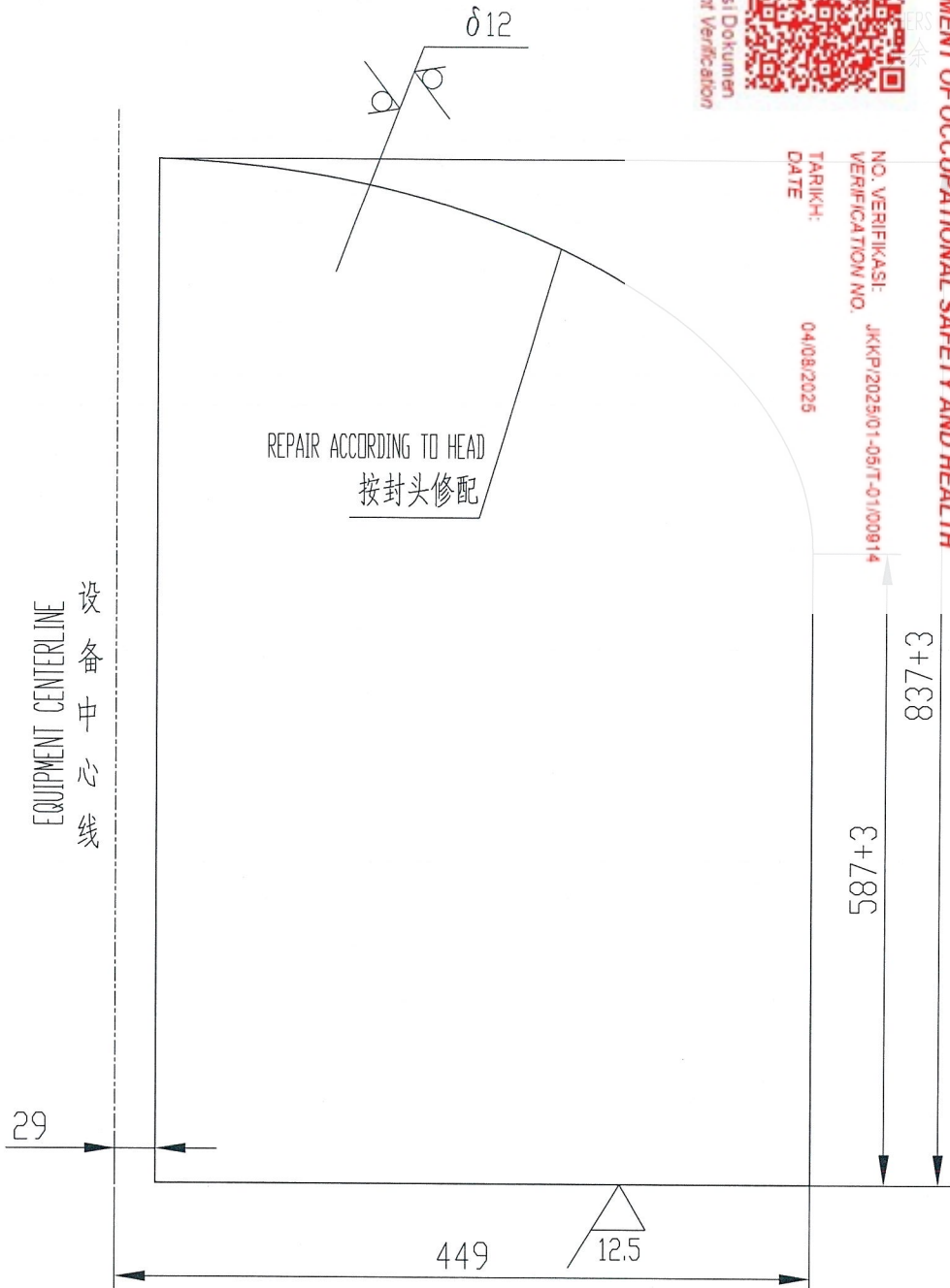
2		STEEL PIPE 钢管Ø168.3X10.97	1	SA-106M Gr.B		12.5	
1		STEEL PIPE 钢管Ø168.3X10.97	1	SA-106M Gr.B		3.2	
NO. 序号	DRAWING NO. 代号	DESCRIPTION 名称	QTY 数量	MATERIAL 材料	UNIT 单重	TOTAL 总重	NOTE 备注
					WEIGHT 重量		
				INLE PIPE 进水管		TXNX23-020.01.08	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日		
DESIGNED BY 设计	陈伟		2024.6.14		MATERIAL 材料	QTY 数量	WEIGHT 重量
CHECKED BY 校对	侯好		2024.6.14			1	15.7
REVIEWED BY 审核							1:4
APPROVED BY 批准					PAGE OF 共 张 第 张		TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司



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TARIKH: 04/08/2025
DATE

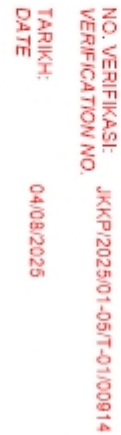


TÜVRheinland®

☐ inspected ☒ reviewed ☐ witnessed

2024.6.18

						PASS PARTITION 分程隔板				TXNX23-020.01.09	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日						
DESIGNED BY 设计	陈伟		2024.6.14			MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	
CHECKED BY 校对	徐子		2024.6.14			SA-516M Gr.485	4	32	1:5		
REVIEWED BY 审核						PAGE OF					
APPROVED BY 批准						共 张 第 张					



45°



2024. 6-18

2		STEEL PIPE 钢管Ø168.3X10.97		1	SA-106M Gr.B			12.5	
1		STEEL PIPE 钢管Ø168.3X10.97		1	SA-106M Gr.B			3.2	
NO. 序号	DRAWING NO. 代号	DESCRIPTION 名 称	QTY 数量	MATERIAL 材料	UNIT 单重	TOTAL 总重	NOTE 备注		
					WEIGHT 重 量				
				OUTLET PIPE 出水管			TXNX23-020.01.10		
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日				
DESIGNED BY 设计	陈伟		2024.6.14						
CHECKED BY 校对	陈伟		2024.6.14						
REVIEWED BY 审核									
APPROVED BY 批准						MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例
							1	15.7	1:4
						PAGE OF 共 张 第 张			
						TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司			

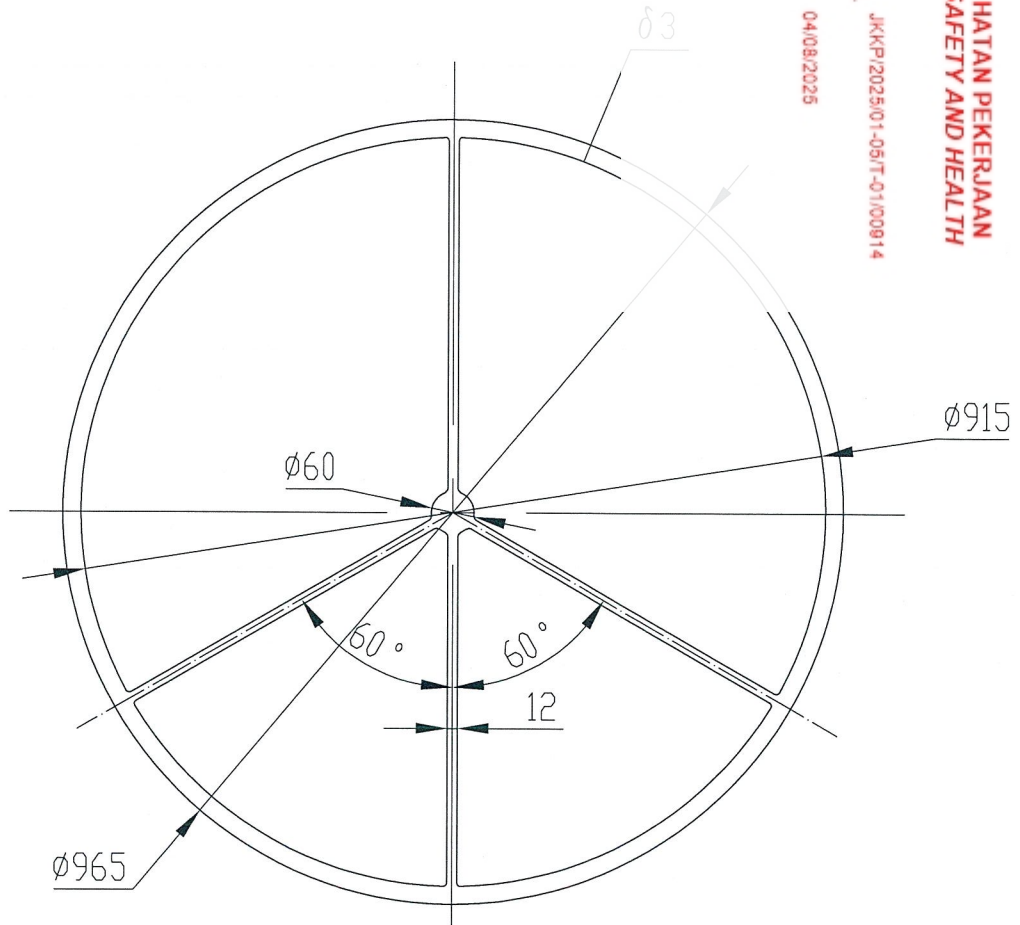


JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



Verifikasi Dokumen
Document Verification

NO. VERIFIKASI: JKKP/2025/01-06/IT-01/00814
VERIFICATION NO.
TARIKH: 04/08/2025
DATE



☐ inspected ☒ reviewed ☐ witnessed

2024.6.18

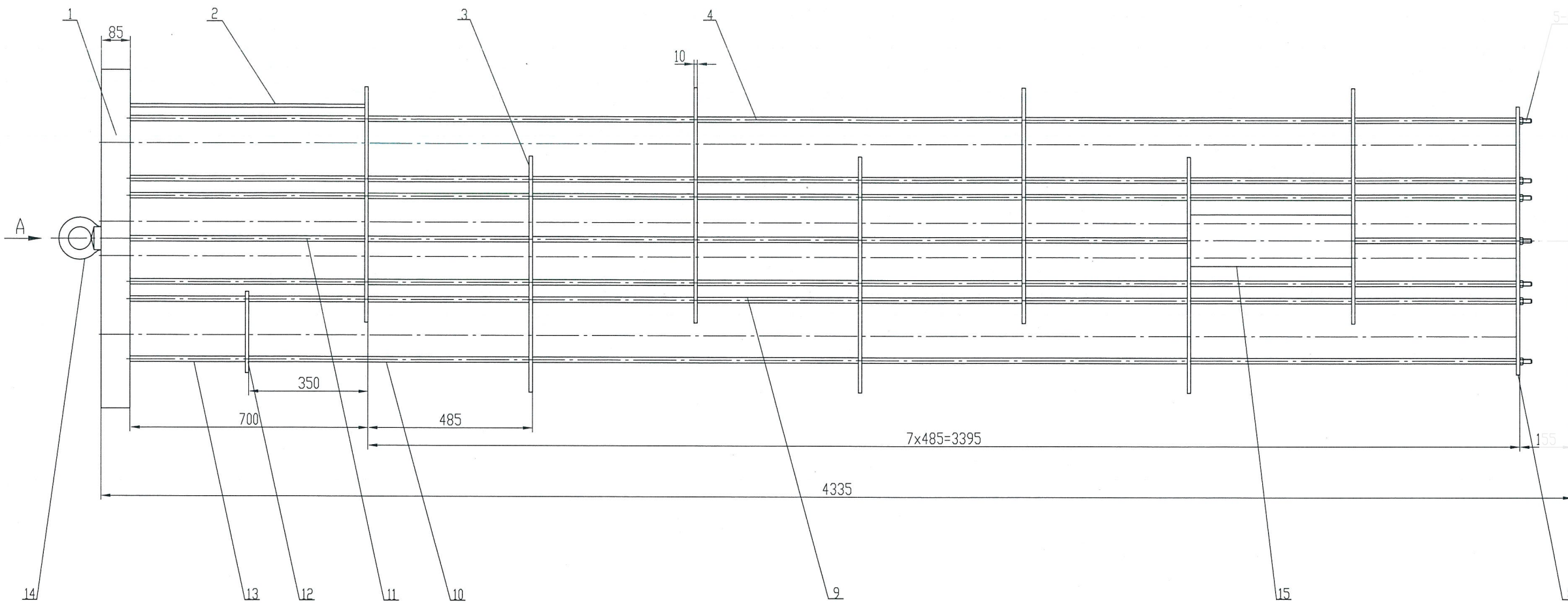
						WATER SIDE GASKET 水侧垫片				TXNX23-020.02	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	 南京汽轮机集团 泰兴宁兴机械有限公司 NANJING TURBINE & ELECTRIC MACHINERY GROUP TAIXING NINGXING MACHINERY CO., LTD.	
DESIGNED BY 设计	陈伟		2024.6.14		PTFE	1	0.2	1:10			
CHECKED BY 校对	陈伟		2024.6.14								
REVIEWED BY 审核											
APPROVED BY 批准					PAGE OF 共 张 第 张						



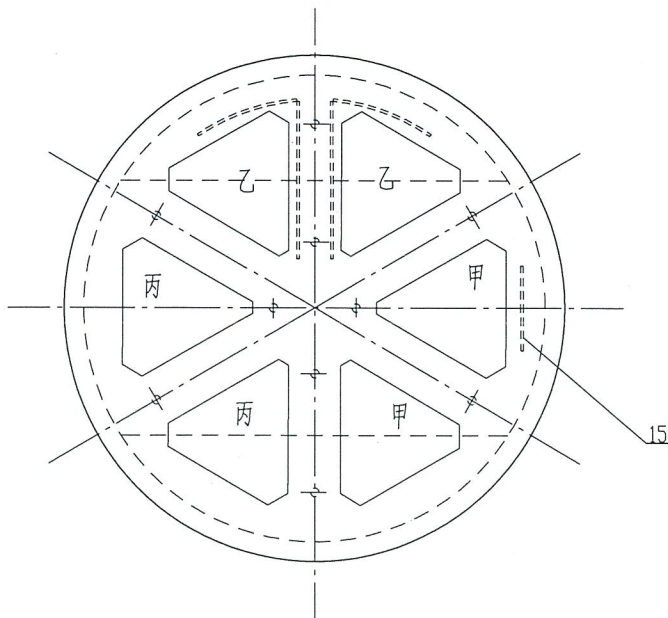
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VERIFICATION NO.
TARIKH: 04/08/2025
DATE

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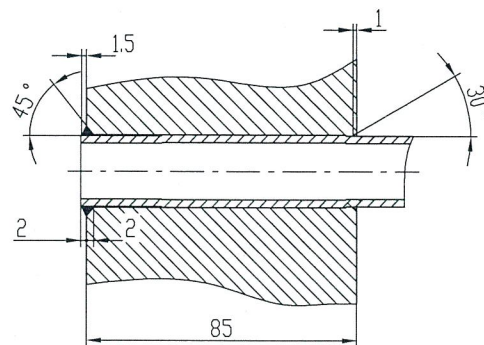
- 1.TUBE BUNDLES SHALL BE MANUFACTURED IN ACCORDANCE WITH ASME SECTION VIII;
管束应按ASME第VIII卷有关制造方面的要求制造;
2.TUBE AND TUBE PLATE ARE CONNECTED BY WELDING;
换热管与管板采用焊接的方式连接;
3.TIGHTEN THE TIE ROD NUTS AND SPOT WELD THEM FIRMLY;
拉杆螺母拧紧后点焊牢固;
4.AFTER COMPLETION OF TUBE BUNDLES, IT IS NECESSARY TO REMOVE IRON FILINGS AND DEBRIS INSIDE AND OUTSIDE TUBES, AS WELL AS OTHER PARTS OF TUBE BUNDLES;
管系完工后需清除管内外及管系其他部分铁屑杂物;
5.FOR OTHER TECHNICAL REQUIREMENTS, REFER TO GA DRAWING.
其他技术要求见总图.



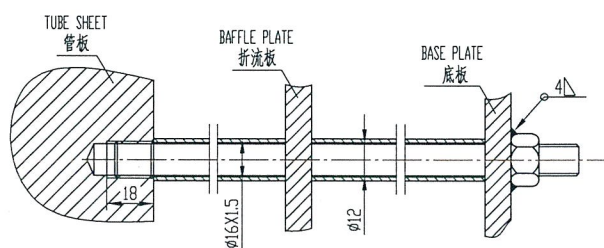
VIEW A
A向



DETAIL DRAWING OF TUBE WELDING
换热管焊接详图



TIE ROD DRAWING
拉杆结构图

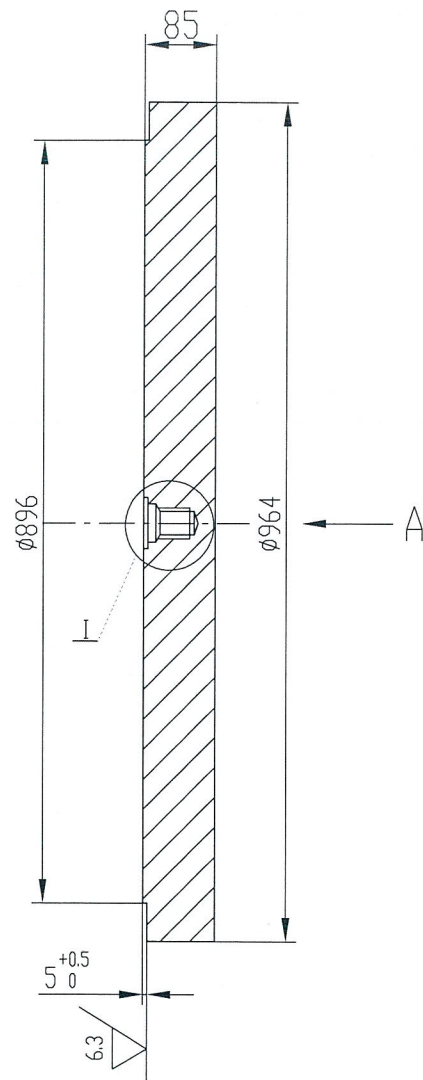


Inspected ☒ reviewed ☒ witnessed ☒

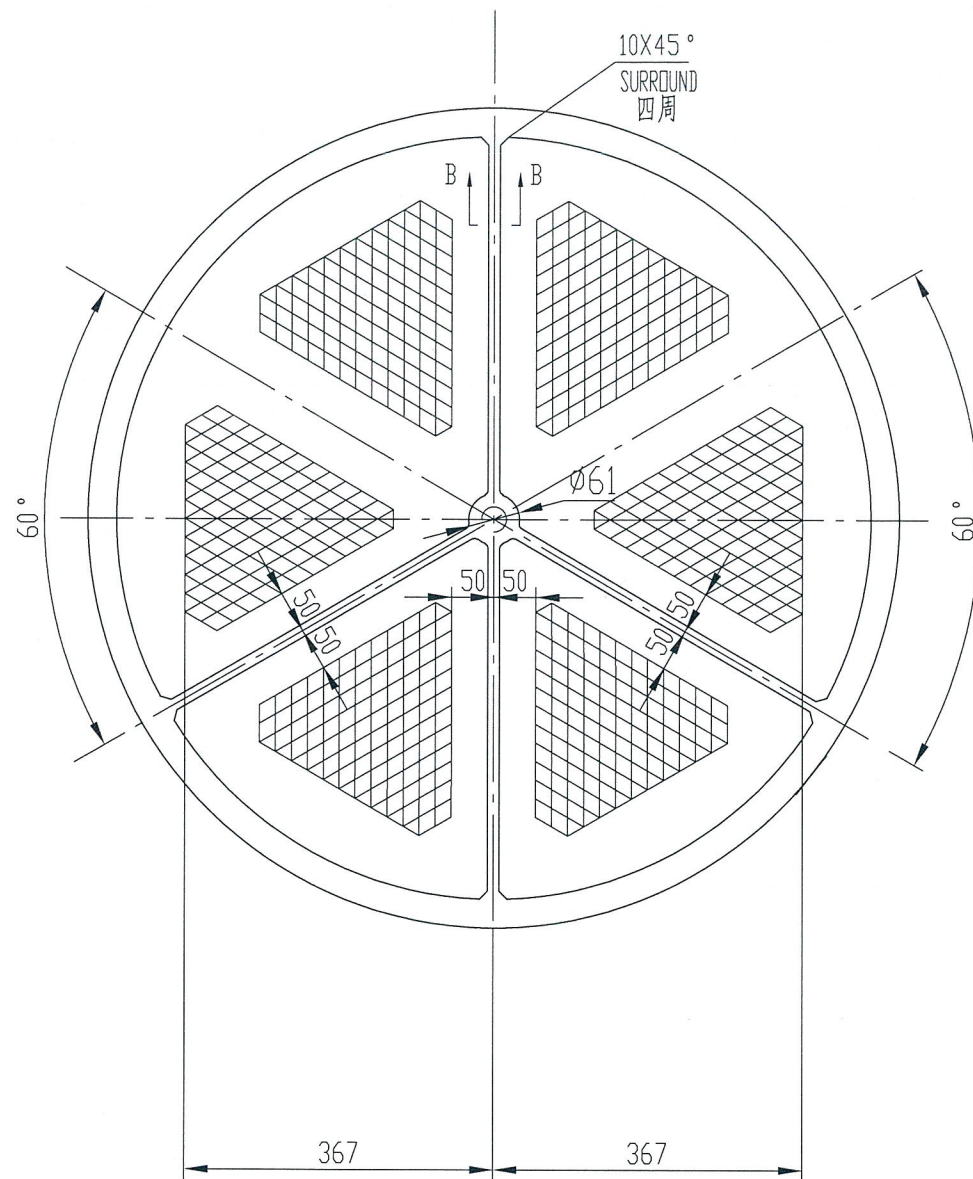
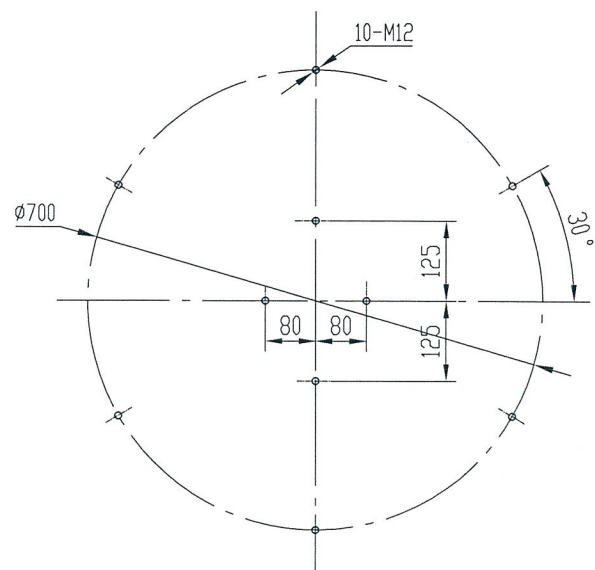
2024.6.18

15		DRAIN INLET BAFFLE 疏水入口折板 475x150x6	1	S30408		3.8	
14	GB/T 825-1988	SCREW 螺栓M36	1	35		2	
13		SPACER 定距管 Ø16X1.5 L=340	5	20	0.18	0.9	
12	TXNX23-021.01.04	PASS PARTITION 隔板	1	Q235B		12	BORROW 借用
11		SPACER 定距管 Ø16X1.5 L=690	7	20	0.37	2.59	
10		SPACER 定距管 Ø16X1.5 L=825	1	20		0.44	
9		SPACER 定距管 Ø16X1.5 L=475	57	20	0.26	14.82	
8	TXNX23-020.03.05	BASE PLATE 底板	1	Q235B		30	BORROW 借用
7	TXNX23-023.01.02	U-TUBE U形管	339	SA-688M TP304		1041	
6	GB/T 6170	NUT 螺母M12	10	8	0.01	0.1	
5	TXNX23-023.01.01	TIE ROD 拉杆	10	Q235B	3.68	36.8	
4		SPACER 定距管 Ø16X1.5 L=960	6	20	0.52	3.12	
3	TXNX23-020.03.02	BAFFLE PLATE 折流板	7	Q235B	37	259	BORROW 借用
2	TXNX23-021.01.01	INLET BAFFLE 进口折板	1	SA-240M 304		4.84	BORROW 借用
1	TXNX23-020.03.01	TUBE SHEET 管板	1	SA-266M Gr.2		357	BORROW 借用
NO. 序号	DRAWING NO. OR STANDARD 图号或标准	DESCRIPTION 名称	QTY 数量	MATERIAL 材料	UNIT 单位	TOTAL 总重	NOTE 备注

TUBE BUNDLES 管束				TXNX23-023.01			
NO. 序号	QTY 数量	WEIGHT 重量	PROPORTION 比例				
1	1	1765	1:7				
PAGE OF 共 页 第 页							



TIE ROD POSITION AT VIEW A
A向拉杆孔位置

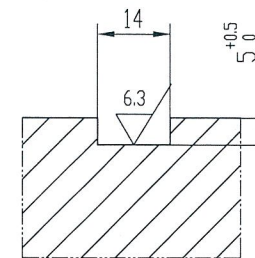


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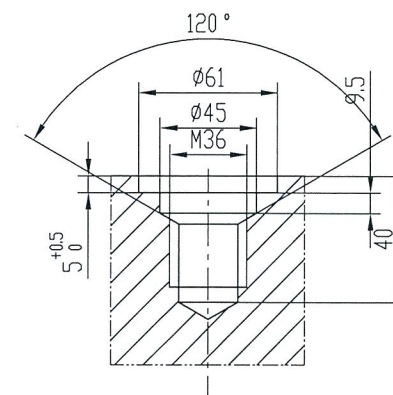
技术要求

- 1.TUBE SHEET MATERIALS SHALL BE SUPPLIED IN A NORMALIZED STATE AND SHALL COMPLY WITH SA266M SPECIFICATION. 管板材料以正火状态供货,且应符合ASME第II卷A篇SA266M规范的所有相关要求;
- 2.TUBE SHEETS SHALL BE MANUFACTURED IN ACCORDANCE WITH ASME SECTION VIII; 管板应按照ASME第VIII卷有关制造方面的要求制造;
- 3.THE ALLOWABLE DEVIATION OF CENTER DISTANCE OF TWO ADJACENT TUBE HOLES IS $\pm 0.3\text{MM}$. THE ALLOWABLE DEVIATION OF ANY TWO TUBE HOLES IS $\pm 1\text{MM}$. SPOT CHECK TUBE HOLES WITHIN CENTRAL ANGLE AREA WHICH IS NOT LESS THAN 96% OF THE TOTAL AREA. TUBE HOLES IS ALLOWABLE. THE VALUE CAN BE 0.025MM LARGER THAN THE DRAWING VALUE AT THE SAME TIME, CHECK THAT THE QUALITY OF THE TUBE HOLES IS NOT LESS THAN 96%. CONTROL THE NUMBER OF LIGAMENT WITH $B_{\text{MIN}} \geq 3.45\text{MM}$ WITHIN 4% OF THE TOTAL AREA. 相邻两管孔中心距允许偏差 $\pm 0.3\text{MM}$,任意两孔中心距允许偏差 $\pm 1\text{MM}$;应抽查不小于60°管板中心角区域内的管孔在这一区域内允许有4%的管孔上偏差,同时抽查此区域内的孔桥宽度 $B \geq 4.7\text{MM}$ 的合格率不小于96%, $B_{\text{MIN}} \geq 3.45\text{MM}$ 的孔桥数控制在4%内;
- 4.THE TUBE HOLE SURFACES SHALL BE FREE FROM DEFECTS SUCH AS LONGITUDINAL AND SPIRAL GROOVES. 管孔表面不允许有贯通的纵向和螺旋向刻痕等缺陷;
- 5.AFTER COMPLETION OF TUBE SHEET, ALL SURFACE BURRS, IRON FILINGS, AND OTHER DEBRIS SHALL BE REMOVED. 管板完工后须清除所有表面的毛刺、铁屑等杂物。

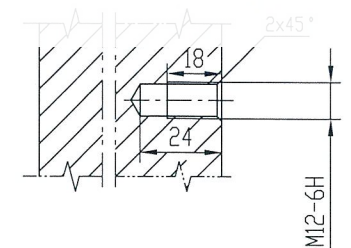
B-B
NO SCALE
不按比例



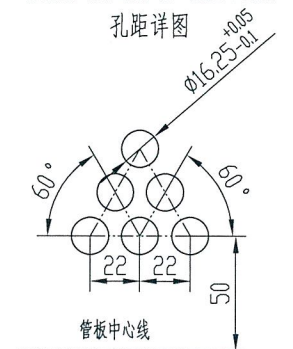
I
NO SCALE
不按比例



DETAIL DRAWING OF TIE ROD HOLES
拉杆孔详图
NO SCALE
不按比例



DETAIL DRAWING OF TUBE PITCH
孔距详图



TÜVRheinland
Inspected ☒ Reviewed ☒ Witnessed ☐
2024.6.18

						TUBE SHEET 管板				TXNX23-020.03.01					
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日										
DESIGNED BY 设计	陈伟		2024.6.14		MATERIAL 材料		QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司					
CHECKED BY 校对	陈伟		2024.6.14		SA-266M Gr.2		1	357	1:5						
REVIEWED BY 审核															
APPROVED BY 批准															
						PAGE OF 共 张 第 张									

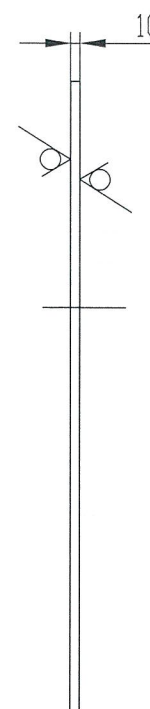
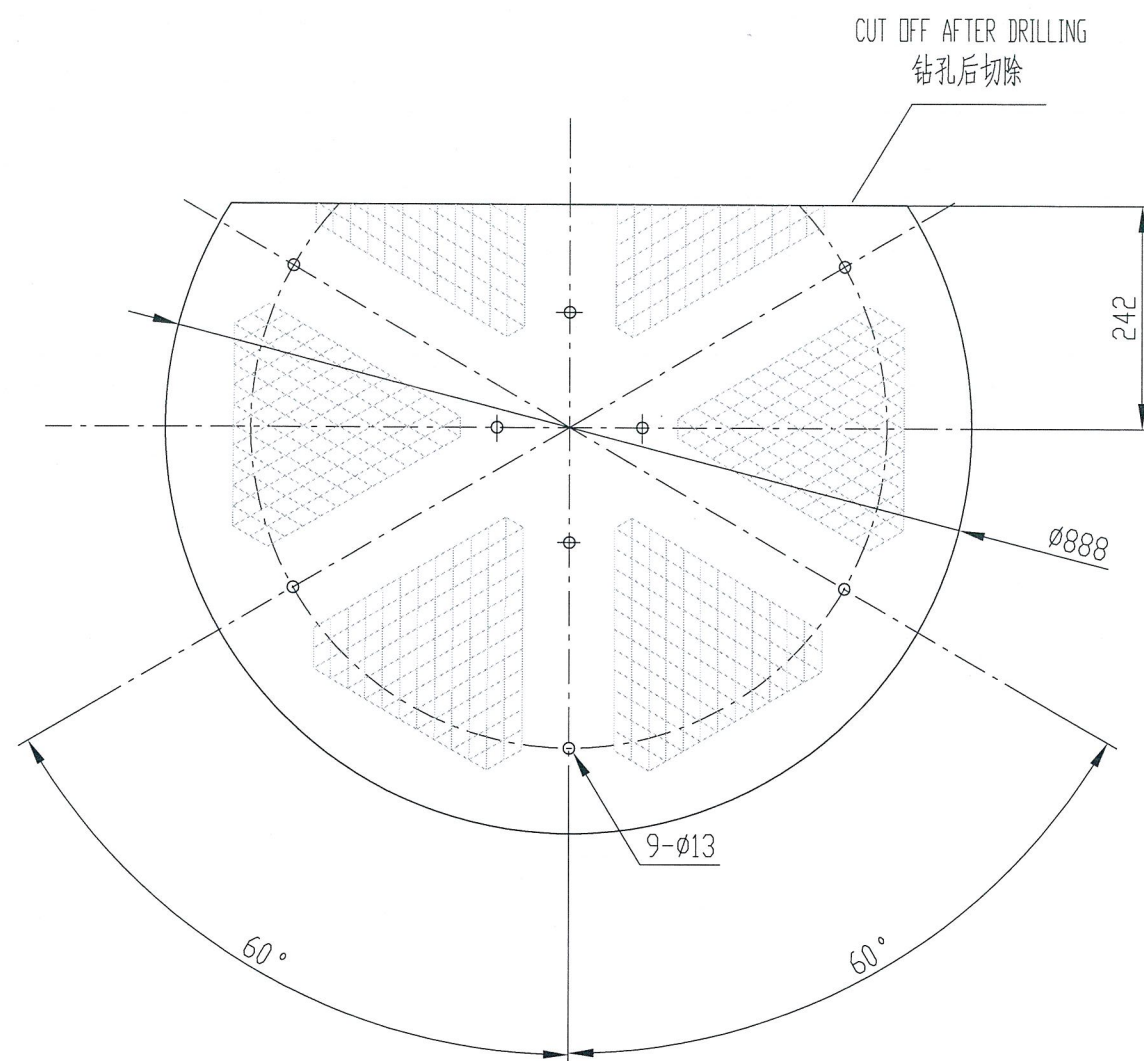


Verifikasi Dokumen
Document Verification

OTHERS
NO. VERIFIKASI: JKKP/2025/01-05/T-01/00914
VERIFICATION NO.
TARIKH: 04/08/2025
DATE



JABATAN KESELAMATAN DAN KESIHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



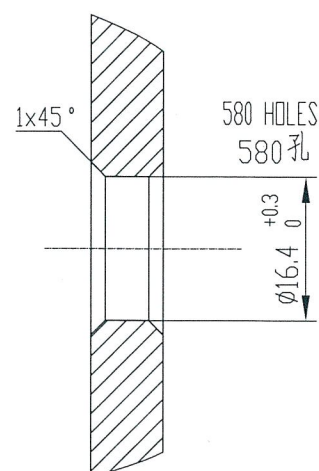
NOTES

技术要求

- 1.THE MARKING OF TUBE HOLES SHALL FOLLOW THE DETAIL OF TUBE SHEET MARKING. EACH TUBE HOLE SHALL BE DRILLED CONCENTRIC WITH THE CORRESPONDING HOLE ON TUBE SHEET.
管孔划线按管板划线详图,各管孔须与管板上对应管孔同心配钻;
- 2.CLEARLY MARK THE GAP POSITION DURING ASSEMBLY;
缺口位置装配时作出明显标记;
- 3.THE BAFFLE PLATE SHALL BE LEVELED BEFORE DRILLING;
折流板钻孔前应校平;
- 4.AFTER DRILLING, BURRS AROUND TUBE HOLES SHALL BE REMOVED.
钻孔后应除去管孔周边毛刺。

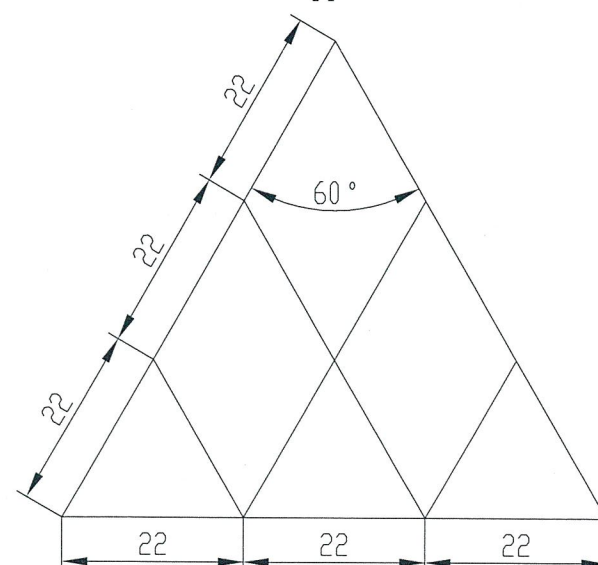
DETAIL DRAWING OF DRILLING
钻孔详图

1:1



DETAIL DRAWING OF TUBE PITCH
孔距详图

1:1



☐ Inspected ☒ reviewed ☐ Witnessed

2024.6.18

						BAFFLE PLATE 折流板				TXNX23-020.03.02	
MARK 标记	QTY 数量	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	
DESIGNED BY 设计						Q235B	6	37	1:8		
CHECKED BY 校对											
REVIEWED BY 审核										PAGE OF 共 张 第 张	
APPROVED BY 批准											



Verifikasi Dokumen
Document Verification

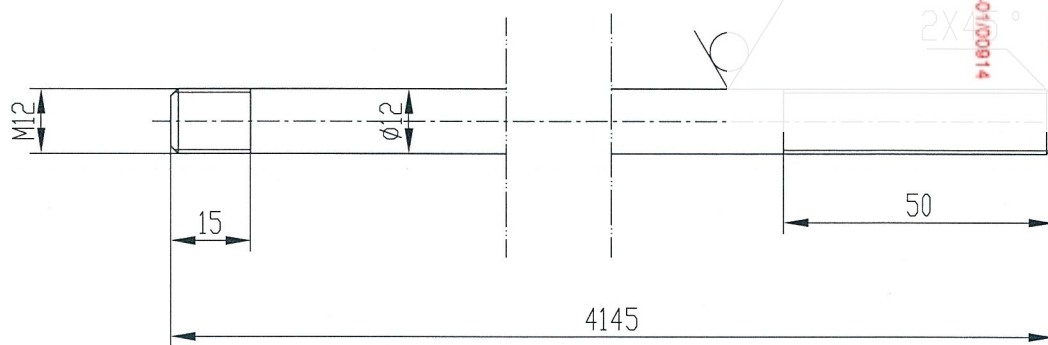
JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



ALIGNMENT IS REQUIRED BEFORE MAKING THREADS

加上螺牙前校直

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VERIFICATION NO.
TARIKH: 04/08/2025
DATE



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Lu Jian
2024.6.18

						TIE ROD 拉 杆				TXNX23-023.01.01		
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司		
DESIGNED BY 设计	陆建		2024.6.14			Q235B	10	3.68				
CHECKED BY 校对	李时		2024.6.14									
REVIEWED BY 审核												
APPROVED BY 批准						PAGE OF 共 张 第 张						

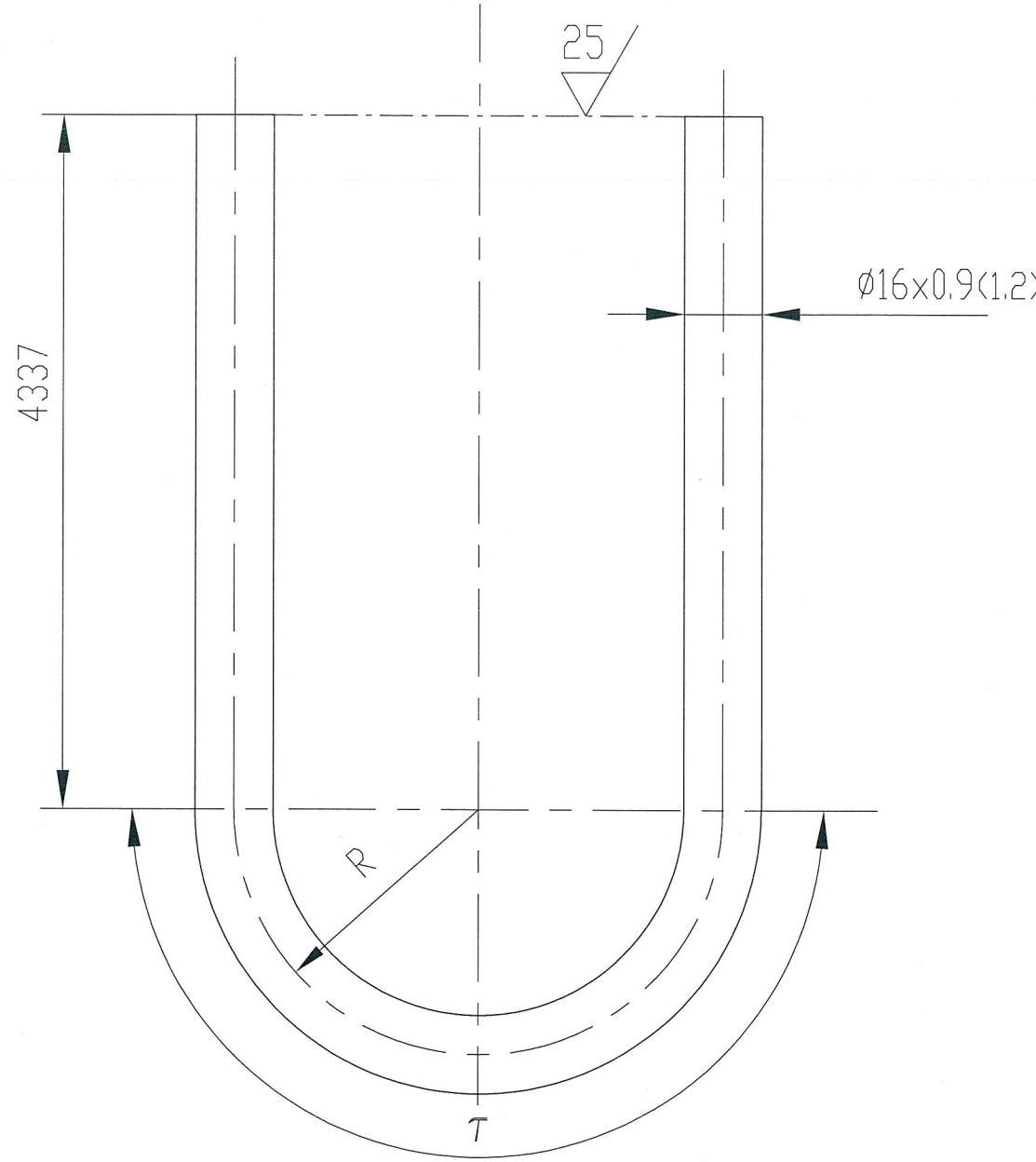


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Document Verification

TABLE OF SIZE AND QUANTITY
尺寸和数量表

NO.	R mm	T mm	TOTAL LENGTH OF TUBE/PC 管子全长/根 mm	TUBE QUANTITY (PC) 管子数量 (根)	TOTAL WEIGHT 总重 (kg)	TOTAL LENGTH 管子总长 mm
1	50	157	8831	36		
2	69	217	8891	39		
3	88	277	8951	39		
4	107	336	9010	36		
5	126	396	9070	33		
6	145	456	9130	30		
7	164	515	9189	27		
8	183	575	9249	24		
9	202	635	9309	21		
10	222	697	9371	18		
11	241	757	9431	15		
12	260	817	9491	12		
13	279	877	9551	9		
TOTAL 合计				339	1041	3086843

NO. VERIFIKASI: JKKP/2025/01-05/T-01/000914
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TARIKH: 04/08/2025
DATE

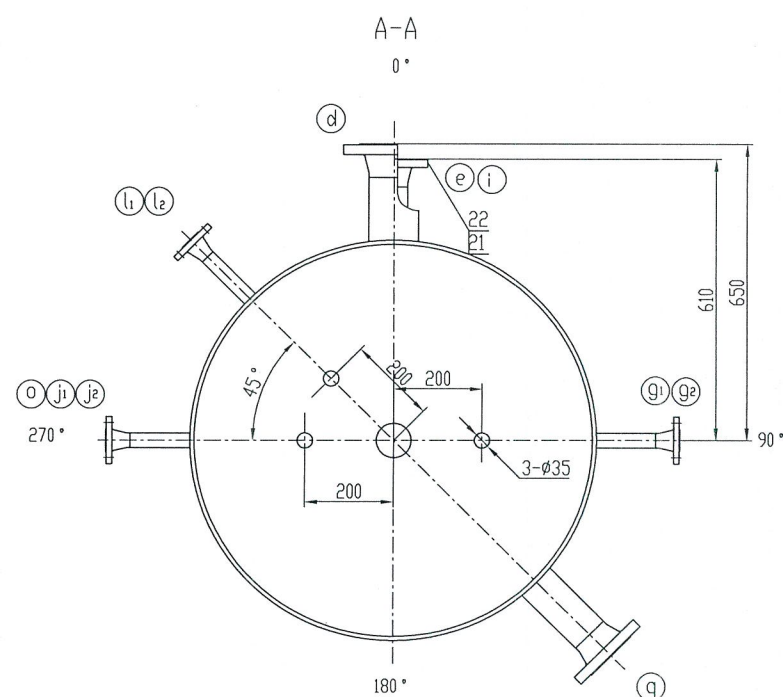
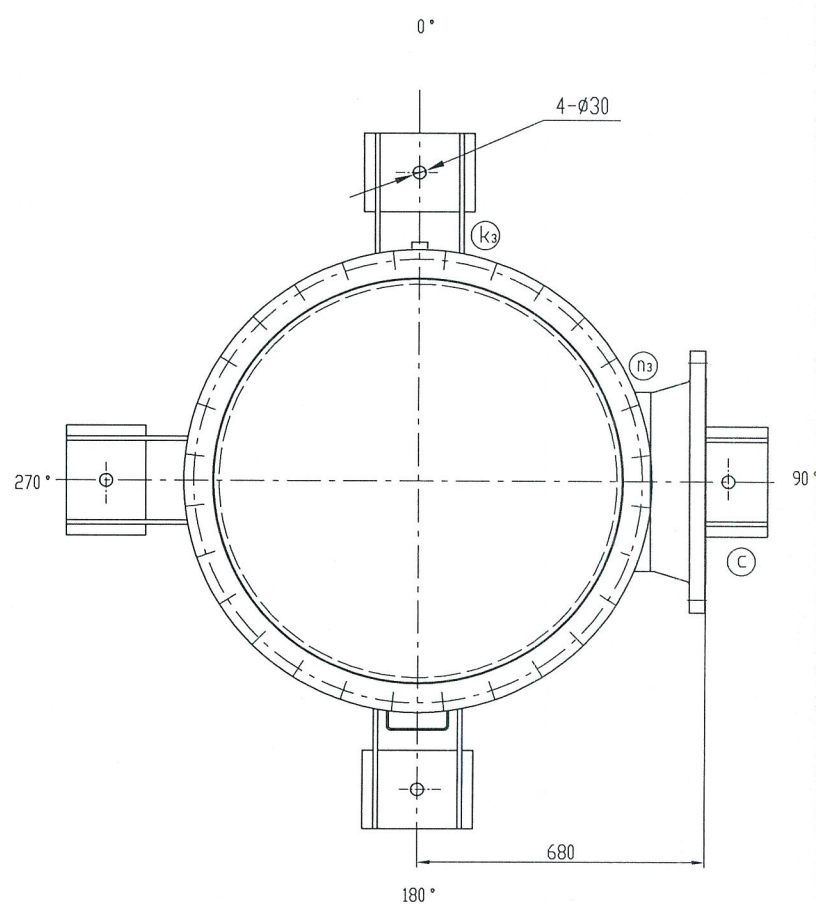
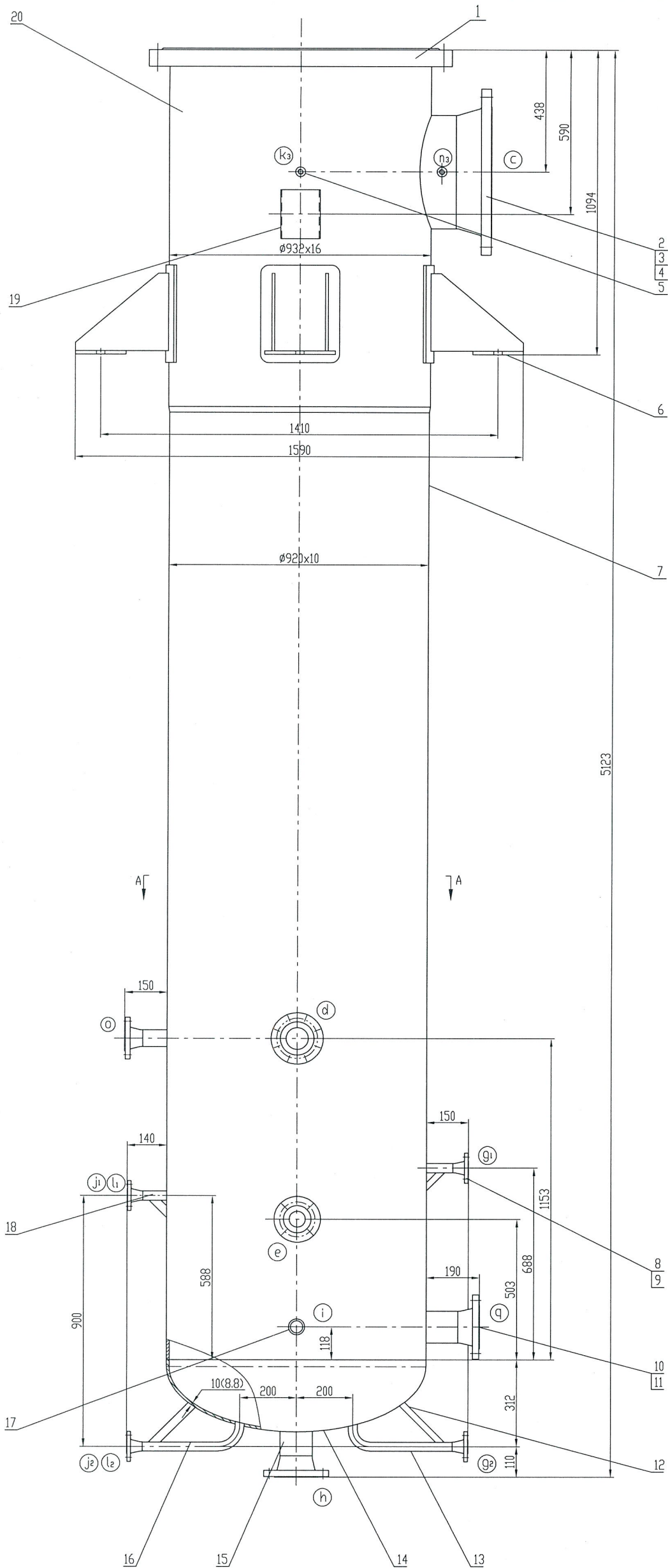


TÜVRheinland®
Inspected ☒ Reviewed ☒ Witnessed ☒
Xi-Jian
2024.6.18

NOTES
技术要求

- 1.TUBES SHALL BE BENT AS A WHOLE. SPLICED WELDING IS NOT ALLOWED;
换热管须整管弯制而成，不得拼焊；
- 2.AFTER BENDING AND FORMING OF TUBES, MAKE HYDROTEST AS PER PIECE WITH A PRESSURE OF 2.75MPa.
KEEP THE PREESURE FOR 10 SECONDS. AFTER THE TEST IS ACCEPTABLE, CLEAN THE WATER STAINS INSIDE TUBES. WHEN THIS REQUIREMENT CANNOT BE MET, THE CHLORIDE ION CONTENT IN WATER SHALL BE CONTROLLED TO NOT EXCEED 25MG/L;
管子弯曲成形后应每根进行水压试验，压力为2.75MPa，保压10S，合格后应将管内水渍处理干净。当无法达到这一要求时，应控制水的氯离子含量不超过25mg/L；
- 3.THE TECHNICAL REQUIREMENTS FOR TUBE PURCHASING SHALL COMPLY WITH ASME SA-688M TP304 STANDARD;
管子订货技术要求按ASME SA-688M TP304标准；
- 4.THE SPECIFICATIONS OF R50 AND R69 TUBES ARE Ø16X1.2.
R50、R69换热管规格为Ø16x1.2。

						TUBE 换热管				TXNX23-023.01.02	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	
DESIGNED BY 设计	陈峰		2024.6.14								
CHECKED BY 校对	徐明		2024.6.14		SA688 TP304	339	1041	1:6			
REVIEWED BY 审核											
APPROVED BY 批准						PAGE OF 共 张 第 张					



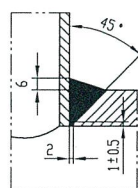
TÜV Rheinland

inspected reviewed witnessed

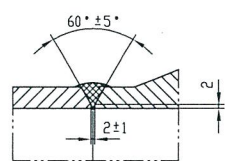
2024.6.18

22	STEEL PIPE	2	SA-106M Gr.B	0.73	1.46	e o
21	FLANGE	2	SA-105M	3.0	6.0	e o
20	SHELL SECTION	1	SA-516M Gr.485	455	H=1257	BORROW
19	NAMEPLATE	1	ASSEMBLY PARTS	3		
18	STEEL PIPE	1	SA-106M Gr.B	0.56	1.2	12
17	STEEL PIPE (INTERNAL EXTENSION 20)	1	SA-182M F304	1.2		i
16	BEND	2	SA-106M Gr.B	1.3	2.6	12
15	NOZZLE	1	SA-106M Gr.B	2.73		h
14	LOWER HEAD	1	SA-516M Gr.485	74.1		BORROW
13	BEND	1	SA-106M Gr.B	1.33		92
12	SUPPORT PLATE	1	SA-516M Gr.485	8.6		LONGH ADAPTATION
11	STEEL PIPE	2	SA-106M Gr.B	2.03	4.06	q d
10	FLANGE	3	SA-105M	5.2	15.6	q d h

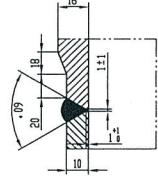
CATEGORY B WELDED JOINT OF NOZZLE
接管B类焊接接头
NO SCALE
不按比例



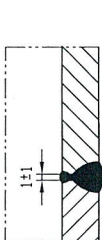
CATEGORY C WELDED JOINT OF NOZZLE-TO-FLANGE
接管与法兰C类焊接接头
NO SCALE
不按比例



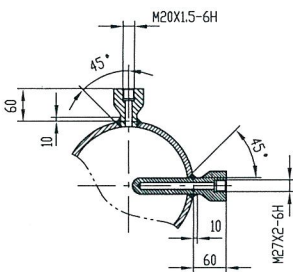
CATEGORY A WELDED JOINT OF NOZZLE-TO-FLANGE
接管与法兰A类焊接接头
NO SCALE
不按比例



CATEGORIES A,B WELDED JOINT
A,B类焊接接头
NO SCALE
不按比例



WELDED JOINTS OF THERMOMETER AND PRESSURE GAUGE
温度计及压力表焊接接头
NO SCALE
不按比例



VERIFICATION NO. JKKP/2025/01-05/T-01/00914
DATE: 04/08/2025
TARIKH: 04/08/2025
NO. VERIFIKASI: JKKP/2025/01-05/T-01/00914
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH
JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
VERIFIKASI DOKUMEN
QR CODE
MUKA SURAT: 1
HALAMAN: 1

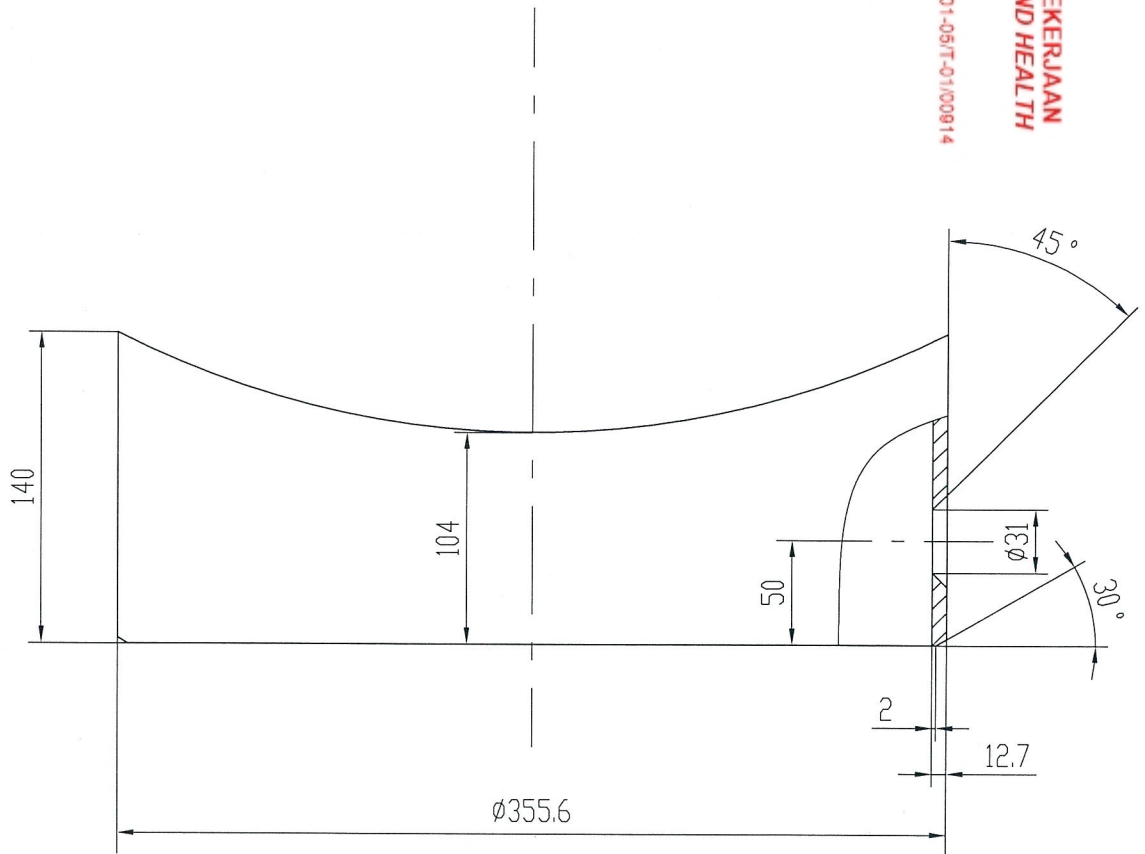


JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
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Verifikasi Dokumen
Document Verification

NO. VERIFIKASI: JKKP/2025/01-06/IT-01/00614
VERIFICATION NO.
TARIKH: 04/08/2025
DATE



TÜVRheinland®

☐ inspected ☒ reviewed ☐ witnessed

Xu Jian
2024.6.18

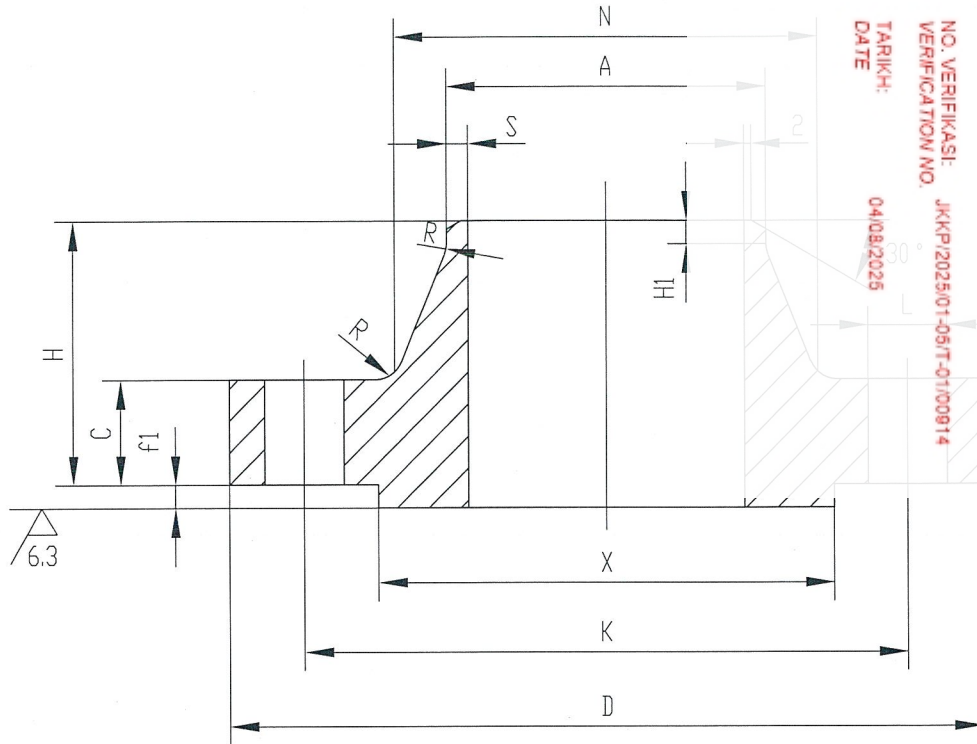
						STEAM INLET PIPE 进汽管				TXNX23-021.02.01					
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日										
DESIGNED BY 设计		陈伟		2024.6.14		MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司					
CHECKED BY 校对		李时		2024.6.14		SA-106M Gr.B	1	7.5	1:2						
REVIEWED BY 审核						PAGE OF 共 张 第 张									
APPROVED BY 批准															



JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



NO. VERIFIKASI: JKKP/2025/01-06/T-01/00614
VERIFICATION NO.
TARIKH: 04/08/2025
DATE



ASME B16.5 WN RF

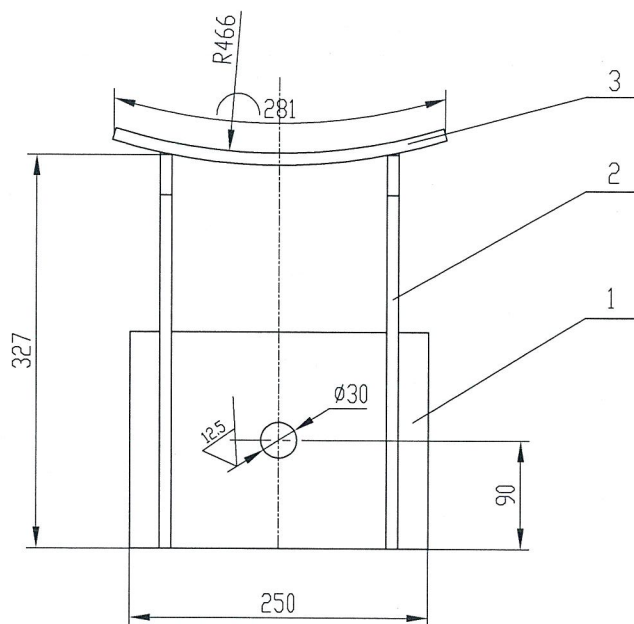
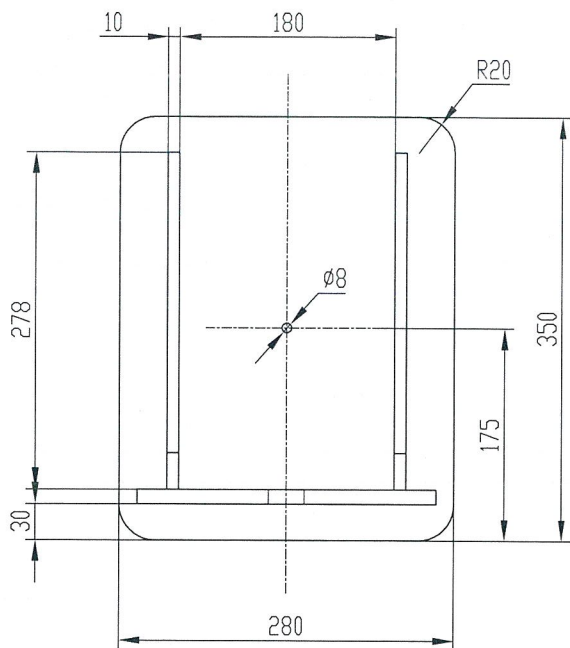
MODEL 型号	A	D	K	L	NO. OF SCREW HOLES 螺孔数	X	f1	C	H	S	N	R	NO. OF FLANGE 法兰数	H1	UNIT WEIGHT 单件重	MATERIAL 材料
class150 1"	33.4	108	79.2	5/8"	4	50.8	1.5	12.7	54	4.55	49	4	6	6	1.0	SA-105M
class150 2"	60.3	152	120.6	3/4"	4	91.9	1.5	17.5	62	5.54	78	8	2	6	3.0	SA-105M
class150 3"	88.9	190	152.4	3/4"	4	127	1.5	22.4	68	7.62	108	10	3	10	5.0	SA-105M
class150 14"	355.6	533	476.2	1-1/8"	12	412.8	1.5	33.3	125	12.7	400	13	1	16	51.7	SA-105M



☐ inspected ☒ reviewed ☐ witnessed

Li. Jian
2024.6.18

						FLANGE 法 兰				TXNX23-021.02.02				
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司				
DESIGNED BY 设计	陈伟		2024.6.14											
CHECKED BY 校对	陈伟		2024.6.14											
REVIEWED BY 审核														
APPROVED BY 批准														
						PAGE OF 共 张 第 张								



Verifikasi Dokumen
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JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
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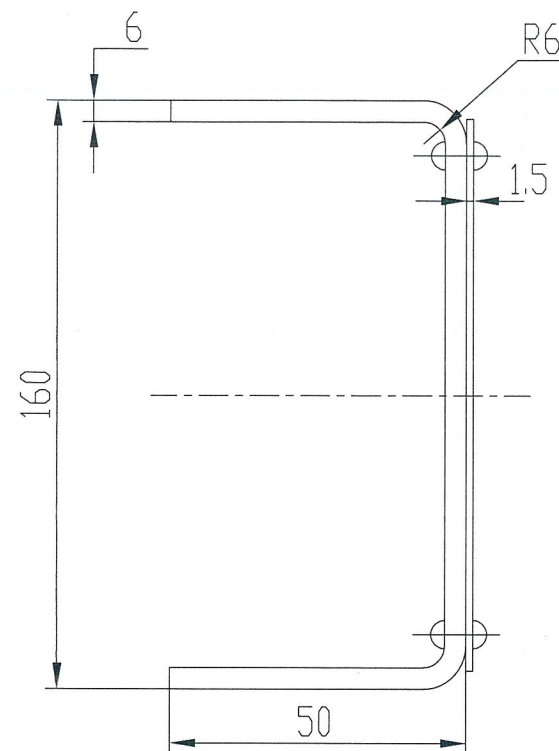
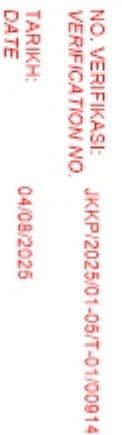


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☐ Inspected ☒ reviewed ☐ witnessed

2024.6.18

3	本图	垫板 $\delta 10$	1	SA-516M Gr.485		7.64	
2	本图	筋板 $\delta 10$	2	Q235B	4.3	8.6	
1	本图	底板 $\delta 12$	1	Q235B		3.45	
PART NO. 件 号	DRAWING NO. OR STANDARD 图号或标准	DESCRIPTION 名 称	QTY 数量	MATERIAL 材 料	UNIT 单	TOTAL 总	NOTE 备 注
					WEIGHT 质量 (KG)		
			EAR LUG 耳式支座		TXNX23-020.05.03		
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日		
DESIGNED BY 设计	陈伟		2024.6.14		MATERIAL 材料	QTY 数量	WEIGHT 重量
CHECKED BY 校对	徐时		2024.6.14		ASSEMBLY PARTS 组合件	4	19.69
REVIEWED BY 审核					PROPORTION 比例	1:4	
APPROVED BY 批准					PAGE OF 共 张 第 张		TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司



说明: NOTES

1. 铭牌由1.5mm的不锈钢制造.

NAMEPLATE SHALL BE MADE OF STAINLESS STEEL WITH THICKNESS OF 1.5MM.

2. 铭牌上的线条、字母和数据框格凸起于板面,且抛光处理,其余部分浸蚀0.1mm深且涂以黑色瓷釉。

LINING, LETTERS, AND BLOCKS OF DATA SHALL BE EMBOSSED AND POLISHED,
OTHER PARTS SHALL BE ENAMEL (0.1MM) DEPTH AND BLACK STOVE ENAMEL PAINTED.

3. 所有必须的标记为5mm高.

ALL THE REQUIRED MARKINGS ARE 5MM HIGH.

- #### 4. 容器竣工后敲认证标记.

THE CERTIFICATION MARK , SHALL BE STAMPED .

3		铭牌架 BRACKET	1	SA-516M.Gr.485		3.0		
2	ISO 1482	铆钉 3.5x 9.5 RIVET	8	SS 304	0.015	0.06		
1		铭牌 $\delta=1.5$ NAMEPLATE	2	SS 304	0.1	0.2		
件 号	图号或标准	名 称	数量	材 料	单 UNIT	总 TOTAL	备 注	
PART NO.	DWG. NO. OR STANDARD	DESCRIPTION	QTY.	MATERIAL	质 量 (kg) WEIGHT		NOTE	
			铭牌 NAMEPLATE		TXNX23-023.02.01			
					版本 REV. NO.	数量 QTY.	重量 WEIGHT	比例 PROPORTION
					0	1	2	-
					泰兴宁兴机械有限公司			
标记 LABEL	处数 PLACE	分区 SUBAREA	更改文件号 NO.OF DOCUMENT	签名 SIGNATURE	年月日 DATE			
设计 DESIGN								
校对 COLLATION								
审核 CHECK								
批准 APPROVE								
共 张 第 张 PAGE OF								